

Chapter 5: Interaction

Joint Interventions, 16 Response Types & Sufficient-Component Causes

Applied Use Case: Retention Offer (A) & Advisory Call (E) on Subscription Churn (Y)

1. Joint Interventions & Counterfactual States (Section 5.1, pp. 57–58)

In real-world CRM and marketing strategy, decisions rarely involve a single isolated stimulus. Instead, companies evaluate **joint interventions** involving two or more active business levers:

- **Retention Discount (A):** $A = 1$ (discount assigned), $A = 0$ (no discount).
- **Advisory Onboarding Call (E):** $E = 1$ (1-on-1 consultation), $E = 0$ (no consultation).
- **Primary Outcome (Y):** $Y = 1$ (customer churn / cancellation), $Y = 0$ (customer retention).

For every individual customer i , there exist **4 potential counterfactual outcomes** corresponding to the 4 joint treatment states:

$$Y_i^{a=1,e=1}, \quad Y_i^{a=1,e=0}, \quad Y_i^{a=0,e=1}, \quad Y_i^{a=0,e=0}$$

Key Conceptual Shift: Interaction vs. Effect Modification

- **Effect Modification (Chapter 4):** V is a *passive baseline attribute* (e.g., customer tenure or age). We do not manipulate V ; we merely observe how the effect of A varies across strata of V .
- **Interaction (Chapter 5):** A and E are *two active joint interventions* that the business can simultaneously control, manipulate, and assign to customers.

2. Additive & Multiplicative Interaction (Section 5.1, pp. 57–58; Technical Point 5.1)

Interaction on the **additive scale** (Risk Difference) occurs when the causal effect of the retention discount (A) in the presence of an advisory call ($E = 1$) differs from its effect in the absence of an advisory call ($E = 0$):

$$\Pr[Y^{a=1,e=1} = 1] - \Pr[Y^{a=0,e=1} = 1] \neq \Pr[Y^{a=1,e=0} = 1] - \Pr[Y^{a=0,e=0} = 1]$$

By algebraic symmetry, this equation is identical to stating that the causal effect of E when $A = 1$ differs from the causal effect of E when $A = 0$. Both levers enjoy equal mathematical status.

- **Superadditivity (Synergy):** The combined reduction in churn from granting both $A = 1$ and $E = 1$ is *greater* than the sum of the individual effects of granting $A = 1$ alone and $E = 1$ alone.
- **Subadditivity (Redundancy):** The combined effect is *less* than the sum of individual effects (e.g., the discount alone retains the customer, making the phone call redundant).

3. Identifiability for Joint Interventions (Section 5.2, pp. 58–60)

To evaluate interaction from observational CRM data, the three identifiability conditions must hold for the **joint intervention** $AE \in \{11, 10, 01, 00\}$:

1. **Joint Exchangeability:** $Y^{a,e} \perp\!\!\!\perp (A, E) \mid L$ for all a, e . No unmeasured confounders U affecting A , E , or Y .
2. **Joint Positivity:** $\Pr[A = a, E = e \mid L = l] > 0$ for all strata l and all a, e .
3. **Joint Consistency:** $Y = Y^{A,E}$ for subjects receiving actual treatment status (A, E) .

4. The 16 Latent Counterfactual Response Types (Section 5.3, pp. 60–62; Table 5.2)

With 2 binary treatments (A, E) and a binary outcome (Y) , each customer possesses a profile of 4 counterfactual outcomes $(Y^{1,1}, Y^{0,1}, Y^{1,0}, Y^{0,0})$. Since each outcome can take 2 values (0 or 1), there are exactly $2^4 = 16$ **latent response types** in the population:

Type	$Y^{1,1}$	$Y^{0,1}$	$Y^{1,0}$	$Y^{0,0}$	Interaction Status & Business Interpretation
1	1	1	1	1	Doomed – Churns regardless of A or E (No interaction)
2–3	1	1	1 / 0	0	Interactive types (Outcome depends on joint combinations)
4	1	1	0	0	Responsive only to E (No interaction between A and E)
5–12	<i>Various joint combinations</i>				Interactive types (Synergistic, antagonistic, or joint-dependent)
13	1	0	1	0	Responsive only to A (No interaction between A and E)
14–15	0	1	0	0 / 1	Interactive types
16	0	0	0	0	Immune – Retained regardless of A or E (No interaction)

Core Theoretical Insight: Additive interaction exists in the overall population if and only if subjects belong to the 10 interactive response types. If the entire population consists solely of the 6 neutral types (Types 1, 4, 6, 11, 13, 16), additive interaction is strictly zero.

5. Sufficient-Component Causes & Rothman’s Pies (Sections 5.4–5.6, pp. 62–68)

While counterfactual models answer **”What happens?”** (*effect estimation*), Rothman’s **sufficient-component cause model** (”causal pies”) answers **”How does it happen?”** (*biological/psychological mechanism*).

Sufficiency Test for Synergism (Fine Point 5.1, p. 64)

Synergism between A and E in the sufficient-cause sense occurs if there exists at least one background constellation U where churn/retention occurs *if and only if* $A = 1$ and $E = 1$ are present together.

Empirical Sufficiency Inequality:

$$\Pr[Y^{a=1,e=1} = 1] - (\Pr[Y^{a=0,e=1} = 1] + \Pr[Y^{a=1,e=0} = 1]) > 0$$

If this condition holds, it guarantees the existence of a synergistic causal pie requiring both $A = 1$ and $E = 1$ simultaneously.